2^{nd} Assignment on Machine learning by using python Programming

1. **What is a dataset in the context of machine learning?**
   a) A single record of information.
   b) A collection of related information or records.
   c) A file containing only qualitative data.
   d) A collection of algorithms.
   Answer: b) A collection of related information or records.
   Explanation: A dataset is defined as a collection of related records where each row represents a record and columns represent attributes.

2. **Which of the following is NOT a type of qualitative data?**
   a) Nominal data
   b) Ordinal data
   c) Binary data
   d) Ratio data
   Answer: d) Ratio data
   Explanation: Ratio data is a type of quantitative data, while nominal, ordinal, and binary data are types of qualitative data.

3. **Which operation cannot be performed on interval-scaled data?**
   a) Addition
   b) Subtraction
   c) Division
   d) Calculation of mean
   Answer: c) Division
   Explanation: Interval data lacks a true zero, so ratios and divisions do not have meaningful interpretations.

4. **What type of attribute is 'model year' in the Auto MPG data set?**
   a) Nominal
   b) Continuous
   c) Discrete
   d) Binary
   Answer: c) Discrete
   Explanation: 'Model year' is discrete because it takes on a finite number of values, such as 1970, 1971, etc.

5. **Which of the following similarity measures is most commonly used for text analysis?**
   a) Euclidean Distance
   b) Jaccard Similarity
   c) Cosine Similarity
   d) Manhattan Distance
   Answer: c) Cosine Similarity
   Explanation: Cosine similarity measures the cosine of the angle between two vectors and is widely used in text analysis and recommendation systems.

6. **What is the range of Pearson Correlation Coefficient?**
   a) [0, 1]
   b) [-1, 1]
   c) [-1, 0]

d) [0, ∞]
Answer: b) [-1, 1]
Explanation: The Pearson correlation coefficient measures linear correlation between two variables and ranges from -1 (perfect negative correlation) to 1 (perfect positive correlation).

7. **Which distance measure accounts for the covariance between dimensions?**
a) Euclidean Distance
b) Manhattan Distance
c) Mahalanobis Distance
d) Minkowski Distance
Answer: c) Mahalanobis Distance
Explanation: Mahalanobis distance accounts for correlations between variables by incorporating the covariance matrix in its calculation.

8. **For which type of data is Hamming Distance most suitable?**
a) Continuous data
b) Binary data
c) High-dimensional data
d) Categorical data
Answer: b) Binary data
Explanation: Hamming distance measures the number of differing positions between two strings and is particularly useful for binary and text data.

9. **Why is feature aggregation performed?**
a) To normalize the data
b) To enhance the interpretability of individual data points
c) To reduce the number of data objects and processing time
d) To identify missing values
Answer: c) To reduce the number of data objects and processing time
Explanation: Aggregating data provides a high-level view and reduces memory consumption and processing time.

10. **Which of the following is an example of feature discretization?**
a) Converting temperatures from Celsius to Fahrenheit
b) Categorizing age into "young," "middle-aged," and "senior"
c) Removing duplicate records
d) Performing one-hot encoding
Answer: b) Categorizing age into "young," "middle-aged," and "senior"
Explanation: Discretization groups continuous data into intervals or categories.

11. **What is the goal of dimensionality reduction?**
a) To increase the number of features
b) To reduce the complexity of a dataset
c) To eliminate duplicates in the dataset
d) To discretize continuous data
Answer: b) To reduce the complexity of a dataset
Explanation: Dimensionality reduction reduces the number of features while retaining meaningful information, reducing complexity.

12. **What is the purpose of feature encoding?**
a) To reduce the dataset size
b) To remove outliers
c) To transform data into a machine-readable format

d) To discretize data
Answer: c) To transform data into a machine-readable format
Explanation: Feature encoding ensures data is in a format understandable by machine learning algorithms.

13. **Which sampling technique ensures a representative sample for predefined groups?**
a) Simple Random Sampling
b) Stratified Sampling
c) Sampling with Replacement
d) Sampling without Replacement
Answer: b) Stratified Sampling
Explanation: Stratified sampling divides the data into predefined groups to ensure representativeness.

14. **Which cross-validation method ensures that each fold has proportional representation of each class?**
a) k-Fold Cross-Validation
b) Leave-One-Out Cross-Validation
c) Stratified k-Fold Cross-Validation
d) Random Split
Answer: c) Stratified k-Fold Cross-Validation
Explanation: Stratified k-Fold Cross-Validation ensures that each fold contains a proportional representation of each class, especially useful for imbalanced datasets.

15. **Which of the following datasets would benefit most from dimensionality reduction?**
a) A dataset with two features
b) A dataset with hundreds of features
c) A dataset with no missing values
d) A dataset with balanced classes
Answer: b) A dataset with hundreds of features
Explanation: High-dimensional datasets benefit from dimensionality reduction to improve efficiency and reduce complexity.

16. **For the Pandas head() method, how many rows are returned by default, if you do not specify it?**
a) 10
b) 5
c) 50
d) All
Answer: b) 5
Explanation: The head() function in Pandas returns the first 5 rows of the DataFrame by default.

17. **What is the correct Pandas function for loading CSV files into a DataFrame?**
a) Read_Csv()
b) Read_csv()
c) read_csv()
d) ReadCSV()
Answer: c) read_csv()
Explanation: The read_csv() function from pandas is used to load data from a CSV file.

18. **What does the command cars.iloc[:, 1:5] return?**
a) Select all the rows with columns from 0 to 4th

b) Select all the rows with columns from 0 to 5th
c) Select all the rows with columns from 1 to 4th
d) Select all the rows with columns from 1 to 5th
Answer: c) Select all the rows with columns from 1 to 4th
Explanation: In pandas, .iloc[] uses zero-based indexing for rows and columns.

19. **Which method from sklearn.preprocessing is used to standardize features by removing the mean and scaling to unit variance?**
a) StandardScaler
b) MinMaxScaler
c) Normalizer
d) RobustScaler
Answer: a) StandardScaler
Explanation: StandardScaler standardizes features by removing the mean and scaling to unit variance.

20. **The process of pickling in Python includes:**
a) Conversion of a list into a datatable
b) Conversion of a byte stream into Python object hierarchy
c) Conversion of a Python object hierarchy into byte stream
d) Conversion of a datatable into a list
Answer: c) Conversion of a Python object hierarchy into byte stream
Explanation: Pickling converts a Python object into a byte stream for storage or transmission.

21. **When using the Pandas dropna() method, what argument allows you to change the original DataFrame instead of returning a new one?**
a) dropna(inplace=True)
b) dropna(keep=True)
c) dropna(original=True)
d) dropna(inplace=False)
Answer: a) dropna(inplace=True)
Explanation: This argument modifies the original DataFrame instead of returning a new one.

22. **What is Scikit-learn?**
a) A machine learning library in Python
b) A data visualization library in Python
c) A natural language processing library in Python
d) A web development framework in Python
Answer: a) A machine learning library in Python
Explanation: Scikit-learn provides tools for supervised and unsupervised learning tasks.

23. **What is a correct method to find relationships between columns in a DataFrame?**
a) df.relation()
b) df.rel()
c) df.correlation()
d) df.corr()
Answer: d) df.corr()
Explanation: df.corr() computes pairwise correlation between columns in a DataFrame.

24. **Which function is used to create a histogram in Seaborn?**
a) sns.scatterplot()
b) sns.histplot()

c) sns.barplot()
d) sns.lineplot()
Answer: b) sns.histplot()
Explanation: sns.histplot() visualizes the distribution of a single variable.

25. **Which function is used to create a pair plot in Seaborn?**
a) sns.boxplot()
b) sns.pairplot()
c) sns.corrplot()
d) sns.relationplot()
Answer: b) sns.pairplot()
Explanation: sns.pairplot() visualizes pairwise relationships between multiple variables.